

Unraveling NYC Yellow Taxi Patterns: An In-depth Exploratory Data Analysis



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Introduction

As one of the most populous cities in the United States, New York City witnesses millions of taxi trips every month. This project aims to conduct a quantitative analysis of the New York City Taxi and Limousine Commission (TLC) trip record data to gain a better understanding of it. Additionally, we aim to provide recommendations that might improve taxi drivers' income.

Data Selection

NYC TLC Dataset

The taxi dataset used in this project covers yellow taxi trip data for the year 2018. It records attributes such as **pick-up and drop-off dates/times, pick-up and drop-off locations, trip distances, itemized fares, rate types, payment types, and driver-reported passenger counts**. Yellow taxis are the taxis that are allowed to respond to street hails from passengers in all five boroughs.

The selected timeline for this analysis will encompass January and February, as well as July and August of 2018, in order to compare taxi activity between winter and summer months.

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- **Taxi Zone Shapefile:** Contains the geometric information for each taxi zone.
- **Taxi Zone Look-Up Table:** A table that lists the TLC taxi zone location IDs, location names, and the corresponding boroughs for each zone.

NYC Weather Dataset

The weather data were downloaded from the National Centers for Environmental Information (NCEI). This dataset contains daily weather observations for Central Park in 2018, including metrics such as maximum/minimum temperature, precipitation, snowfall, and wind speed. We hope this will allow us to study how different weather conditions affect taxi usage in NYC.

For this analysis, we assume that the entire New York City experiences the same daily weather conditions as Central Park and that the information provided by the dataset is accurate.



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Preprocessing

The winter and summer data contain approximately 17 million and 16 million trip records, respectively, and both have 17 columns of attributes.

Feature Engineering

Feature engineering is the process of transforming raw data into useful features to maximize the potential of the data. Below is a brief summary of the steps I took:

- Calculate the duration of each trip using the difference between pickup and drop-off time.
- Identify the month, day of the week, and hour for each trip. Then classify whether the trip is in workday, weekend, or, holiday.
- Identify the pickup and dropoff borough of each trip.
- Calculate the tip percentage using total amount and tip amount.

Furthermore, attributes such as MTA Tax, Extra, and Improvement Surcharge, which are not our primary focus, will be dropped to reduce the data's size and complexity.

Data Cleansing

Records with implausible values or errors were removed after examining the distribution of each attribute and using common sense to ensure the dataset's

consistency and accuracy. Some examples are provided below.

- Pickup/drop-off date and time are strictly within the selected month period.
- Pickup/drop-off Location ID should be within the range of [1, 263].
- Passenger count should at least 1 and less than 7 as the maximum number of passengers allowed by law is 6.
- Trip distance should greater than 0 miles but less than 100 miles.
- Fare amount should be at least \$2.5 but at most \$250.
- Tip percentage are less than 50%
- Trip duration should be more than a minute and less than three hours.

Finally, weather data are merged together with the taxi dataset according to the date of each trip.

Analysis and Visualisation

Geospatial and Time Analysis

In Figure 1, the maps display the total numbers of taxi pickups and drop-offs in New York City during the selected timeline. Darker regions indicate heightened taxi activity. It's evident that **pickups are predominantly concentrated in Manhattan, LaGuardia, and JFK, while drop-offs spread further into the outer boroughs.** Notably, Staten Island sees significantly fewer pickups and drop-offs compared to other boroughs, which aligns with it being the least populated borough.

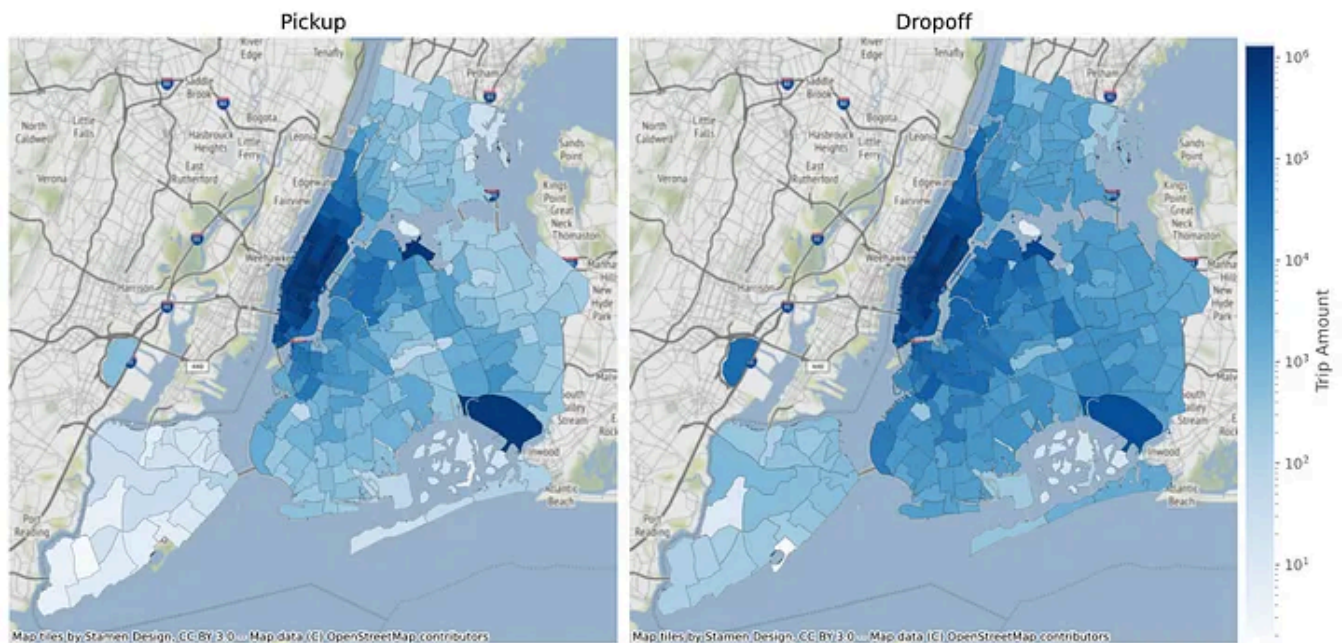


Figure 1. Pick-up vs Drop-off Amount

As illustrated in Figure 2, the evening rush hour during weekdays is noticeably the busiest time. The period from midnight to 6 am sees the lowest pickup frequency on average for weekdays. A distinct pattern emerges when comparing weekdays and weekends at night: the timeframe with the fewest pickups is delayed by approximately two hours on weekends. This shift can likely be attributed to people engaging more in nightlife during the weekend. Another observation of interest is the higher pickup frequency during winter's morning and evening rush hours compared to summer. The cold weather could be a factor, as people might be more inclined to take taxis for their commutes to and from the office.

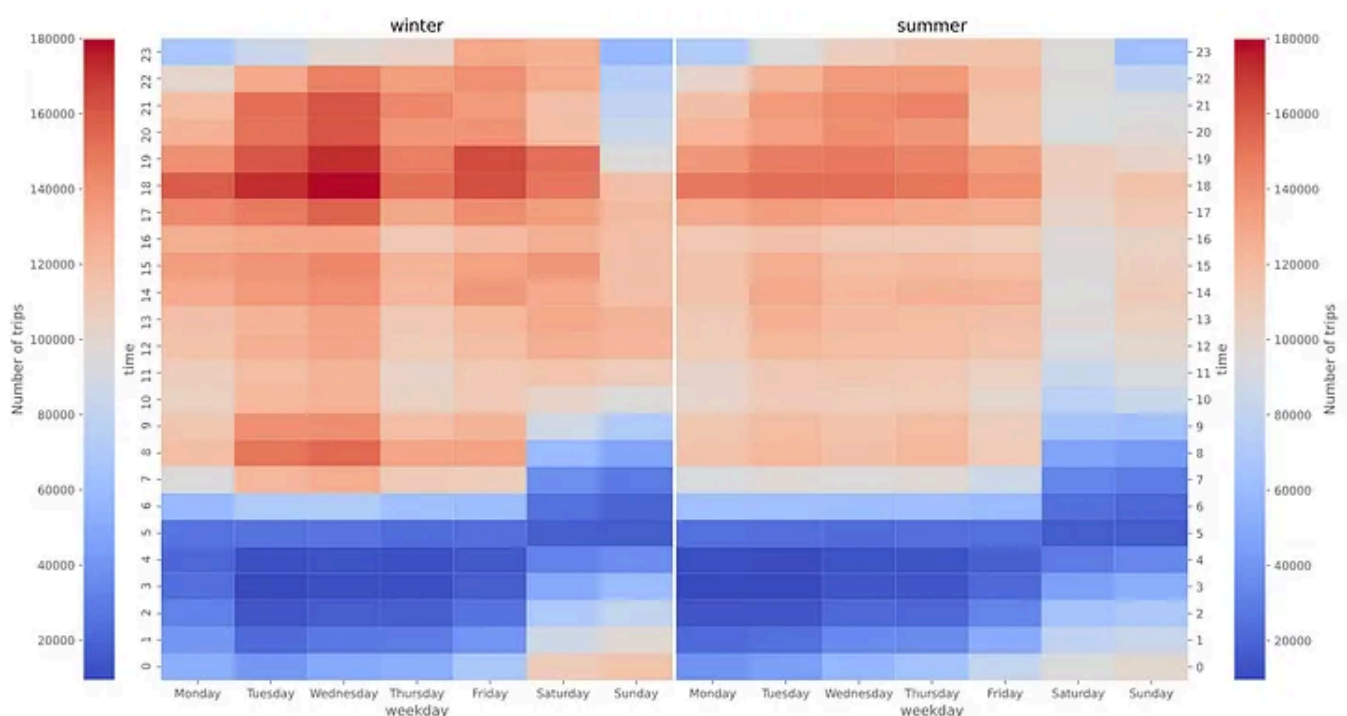


Figure 2. Winter vs Summer Trip Amount Comparison

Weather Impact

Next, we consider the impact of weather. The following scatter plots depict the number of trips taken in relation to various weather conditions. For the bottom two plots, summer trips are excluded as there is no snow during that season.

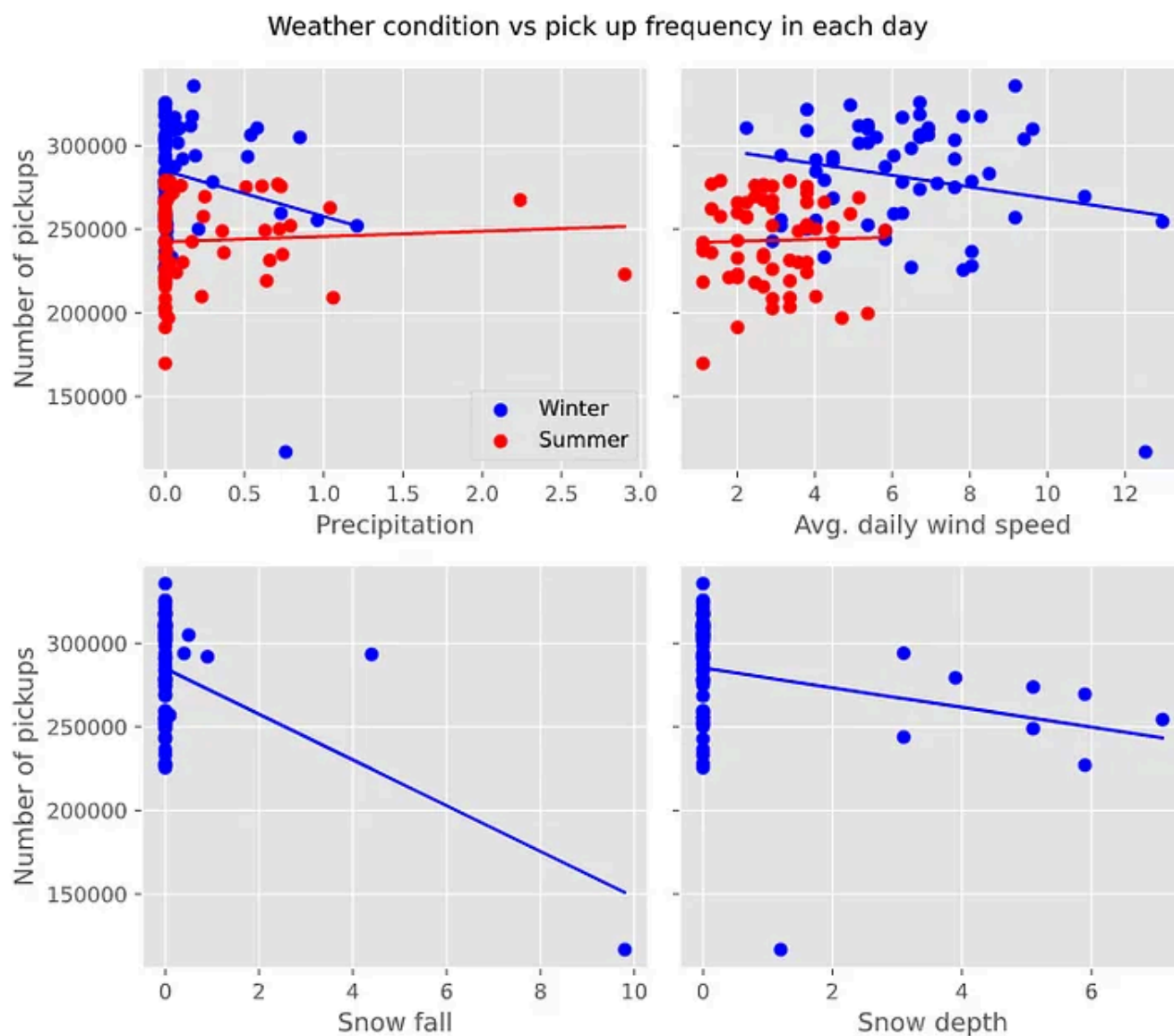


Figure 3. Weather Condition vs Pick-up Frequency

From Figure 3, it's apparent that as precipitation increases during the summer, there's a slight uptick in the daily trip count; however, the trend is reversed in winter. Wind speed doesn't seem to have a significant influence on daily pickup frequency. In contrast, snow depth and snowfall appear to negatively impact daily ridership. Notably, the plot highlights the effect of the January 4, 2018 blizzard in NYC: Central Park reported 9.8 inches of snow, which led to a sharp decline in daily ridership, dropping it to around 120,000.

However, remember that we made an assumption: all taxi zones in NYC experience the same weather conditions as Central Park. This could influence the results of our analysis.

Tipping Behaviour

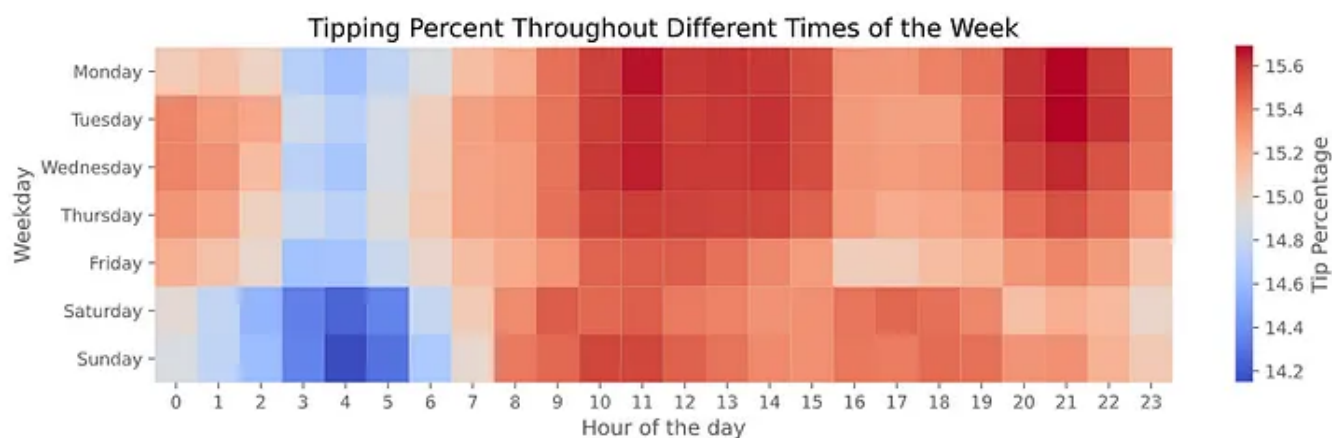


Figure 4. Passenger's Tipping Behaviour Throughout the Week

As illustrated in Figure 4, there's a noticeable difference between weekdays and weekends in tipping patterns, suggesting that passengers tend to tip differently based on the day of the trip. It's also noteworthy that passengers seem to tip a smaller percentage of the total fare in the early morning, specifically from 3 am to 6 am. However, as the day progresses, the average tip percentage steadily increases. There's a slight dip between 16:00 and 19:00 on weekdays, but it reaches its peak with a pronounced surge from 20:00 to 22:00.

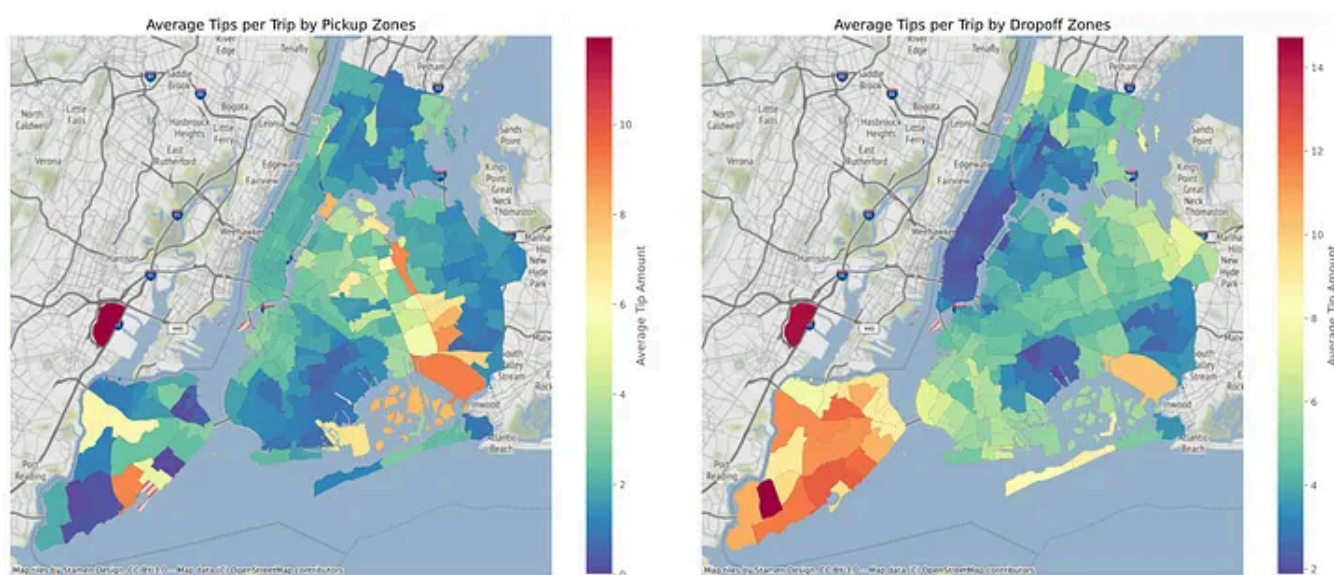


Figure 5. Average Tips/Trip by Pick-up Zone and Drop-off Zone

Figure 5 underscores that, on average, passengers **picked up from Newark Airport, JFK, and Flushing Meadows Corona Park** tend to leave larger tips. In contrast, passengers **dropped off at Great Kills and Oakwood on Staten Island** often tip more generously. This aligns with Staten Island's reputation as one of the more affluent boroughs. However, it's worth noting that some zones have limited trips, so the statistics could be skewed due to an insufficient sample size.

Correlation Between Attributes

Next, we examine the correlation among continuous variables. As illustrated in Figure 6, there's a clear positive relationship among attributes like **"fare amount," "tip amount," "total amount," "trip distance,"** and **"duration"** — this isn't surprising given their nature. Conversely, correlations between other variables aren't as pronounced, with the notable exception of a strong negative correlation between wind speed and daily average temperature.

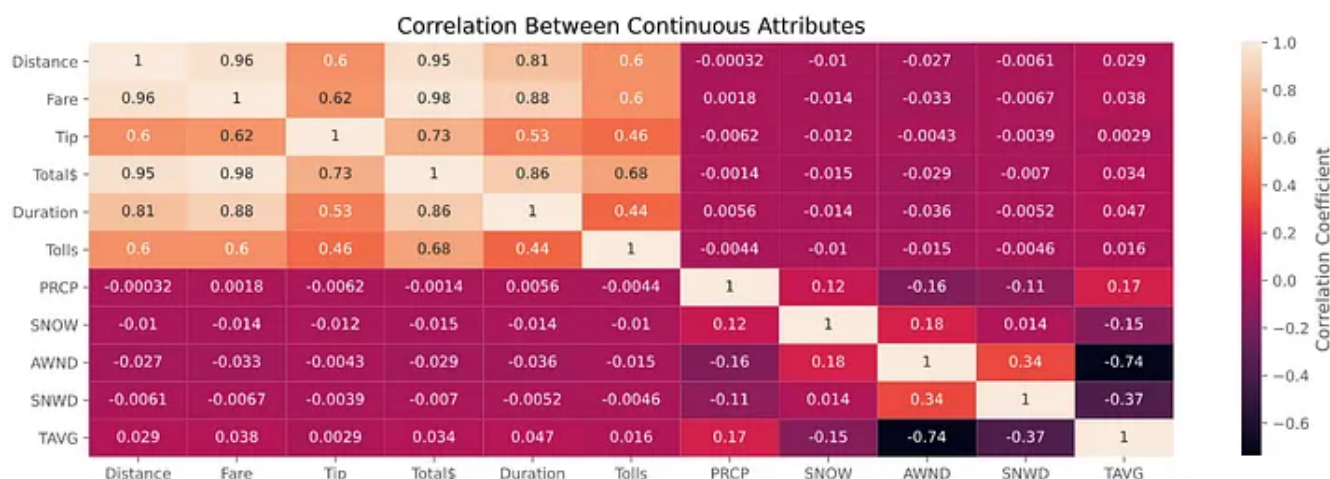


Figure 6. Correlation Between Attributes

Statistical Modelling

Ordinary Least Squares Regression and Gradient Boost Regression

In this section, we'll construct two regression models to forecast the tip amount for relevant taxi trips. We'll evaluate the models' fit using the train-test split method, where the sampled data is divided into an 80% training set and a 20% testing set.

Our initial feature selection is guided by a correlation plot, helping us eliminate redundant predictors and thereby reduce the risk of overfitting. Features, such as weather data, which show no significant correlation with the response variable will

be omitted to prevent obscuring the effects of other predictors. We'll further refine our feature selection using t-tests.

Evaluation and Results

We evaluate the models' performance using the **Root Mean Square Error (RMSE)** and the **Coefficient of Determination (R^2)**. The results of our models are benchmarked against a baseline model, which contains only the intercept and uses the mean as its predicted value.

Goodness-of-fit statistics				
Model	Train Set RMSE	Test Set RMSE	Train Set R^2	Test Set R^2
Baseline	2.342990	2.338763	0	0
Linear Regression	1.294810	1.287552	0.694599	0.696919
GBR	1.300274	1.309557	0.693590	0.690436

Table 1. Model Performance Statistics

Relative to the baseline, our final linear model accounted for an additional 69% of the variance in the tip amount. It achieved an RMSE of approximately 1.3.

Considering the average tip amount is around \$2.59, this indicates that most predictions deviate by about \$1.3 from the actual tip amount. We also experimented with a more advanced gradient boost regression to predict the tip amount. Gradient boosting works by iteratively modeling the residuals to minimize loss. However, its performance did not show significant improvement.

Discussion

In summary, the model's performance was subpar due to its substantial prediction error, rendering it unsuitable for reliable predictions. Diagnostic plots revealed that the residuals aren't normally distributed and exhibit noticeable heteroscedasticity, signaling a breach of the linear model's assumptions. This compromised performance can be attributed to the right-skewed distribution of tip amounts, with a significant portion clustered in the low range between \$0 and \$7.

Recommendations

From the visualization section, several patterns emerge. Weekdays, particularly during evening rush hours, are typically the busiest times. Late-night pickups are more frequent on weekends, likely due to increased late-night activities. When it

comes to tipping, passengers tend to be less generous in the early morning and evening rush hours. However, tip percentages rise during the day, particularly between 10 am to 3 pm, and again at night from 8 pm to 10 pm. Notably, passengers picked up from **Newark and JFK** tend to tip more generously on average. Taxi drivers can leverage these insights to potentially enhance their earnings.

Conclusion

In conclusion, our exploratory analysis of yellow taxi trip records has shed light on the relationships between various attributes. The use of visualization plots proved invaluable, offering us deeper insights into taxi usage patterns both weekly and hourly. We also attempted to build models predicting tip amounts, though their performance was not optimal. Looking ahead, if we can access more detailed weather data, we might be able to more accurately gauge the impact of varying weather conditions on taxi activity.

Feel free to visit my GitHub Repository for this analysis

<https://github.com/greysonchung/New-York-Taxi-Data-Analysis>

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New York Taxi

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Data Analysis

Data Visualisation

Exploratory Data Analysis

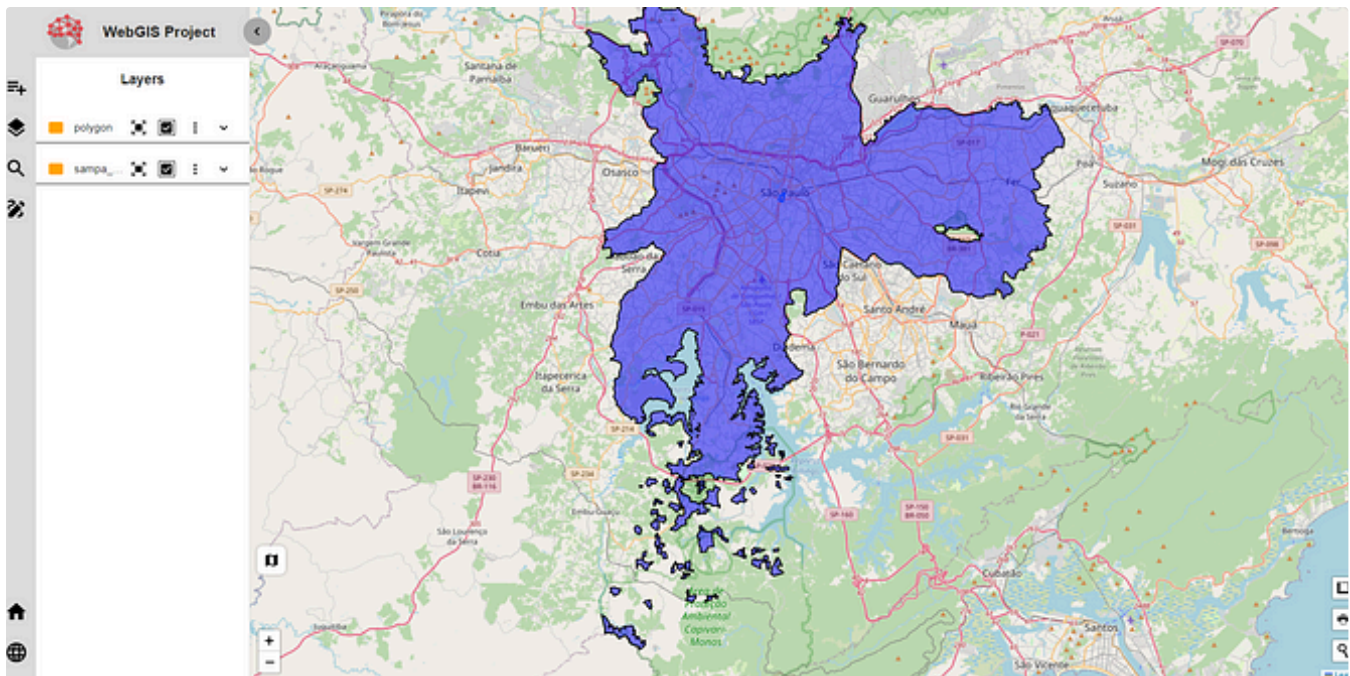


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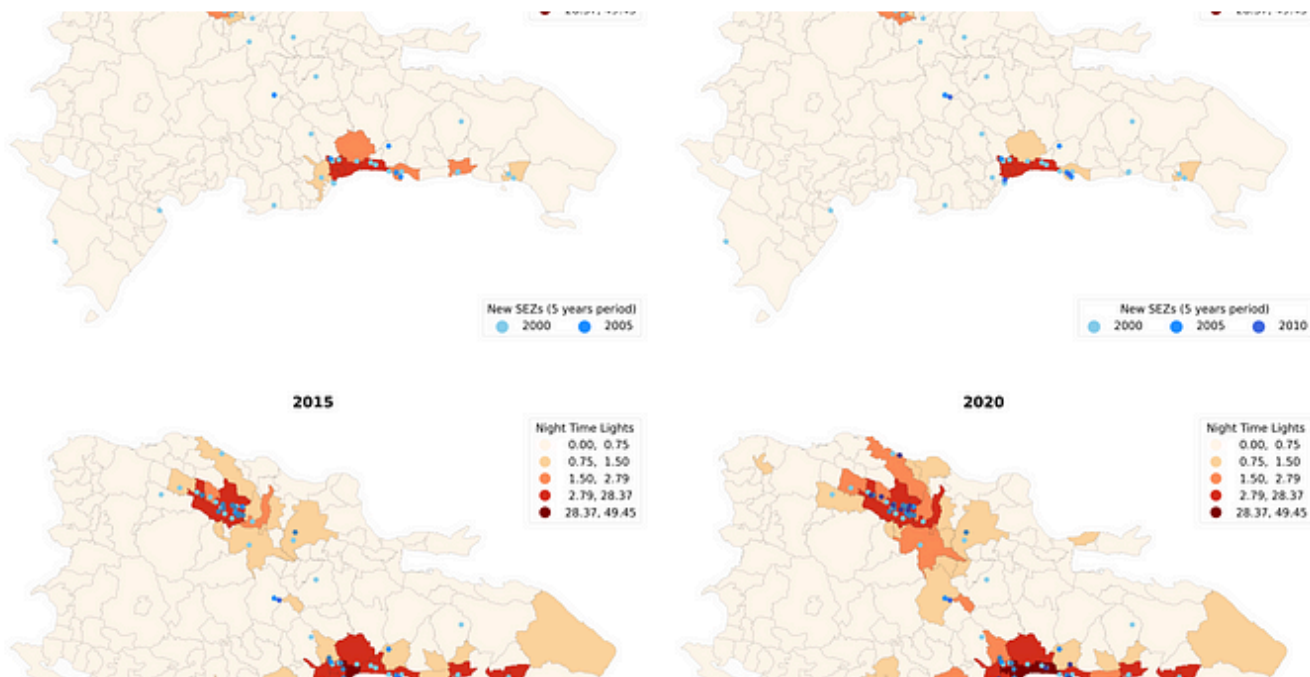


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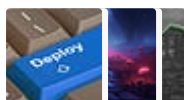
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- Integrated Iframes for credit cards and bank accounts to secure 80% of all consumer traffic and prevent CSRF, cross-site scripting, and cookie-jacking
- Led Your Transactions implementation for JavaScript front-end framework to showcase consumer transactions and reduce call center costs by \$25 Million
- Recovered Saudi Arabia checkout failure impacting 4000+ customers due to incorrect GET form redirection

Projects

NinjaPrep.io (React)

- Platform to offer coding problem practice with built in code editor and written + video solutions in React
- Utilized Nginx to reverse proxy IP address on Digital Ocean hosts
- Developed using Styled-Components for 95% CSS styling to ensure proper CSS scoping
- Implemented Docker with Seccomp to safely run user submitted code with < 2.2s runtime

HeatMap (JavaScript)

- Visualized Google Takeout location data of location history using Google Maps API and Google Maps heatmap code with React
- Included local file system storage to reliably handle 5mb of location history data
- Implemented Express to include routing between pages and jQuery to parse Google Map and implement heatmap overlay



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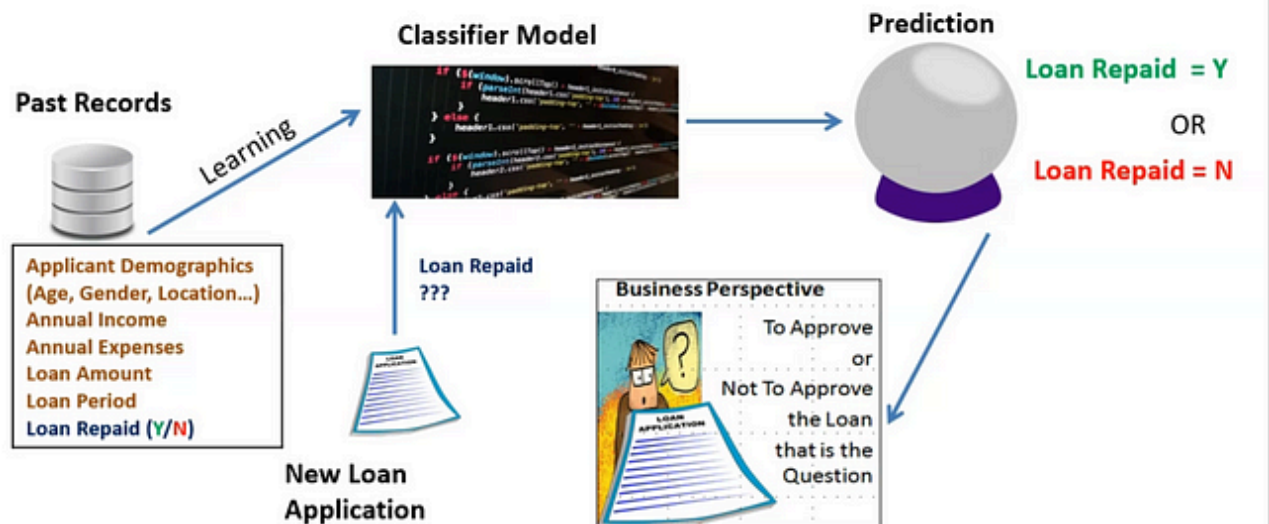
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Data Science Problem - Classification

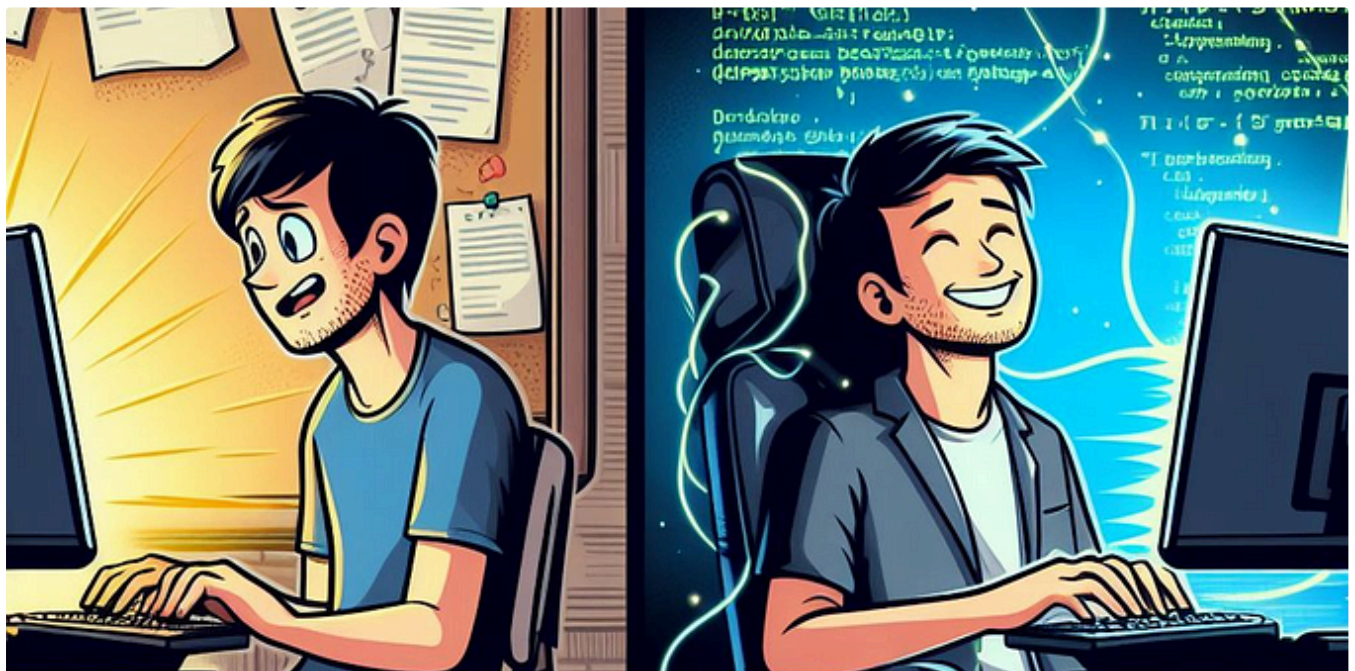


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